



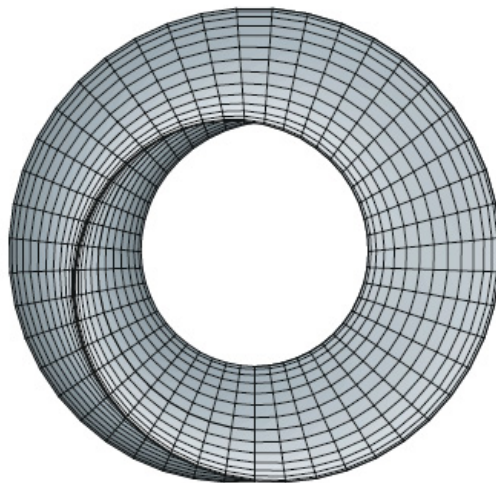
UNIVERSITY OF ZIMBABWE

HMTHCS111 CALCULUS 2

Calculus of Several Variables

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Chapter 2

Vectors and the Geometry of Space

Don't be pushed around by the fears in your mind. Be led by the dreams in your heart.

—Roy T. Bennett, *The Light in the Heart*.

A vector is a quantity that is characterized by **magnitude** and **direction**. We also use the term **length** for magnitude. A scalar, on the other hand, is a quantity which has **magnitude** only. To differentiate the types of quantities, let's consider a typical vector, **displacement** or **change of position**. In order to specify displacement, we need to know two things: *how far?* and *in what direction?* In other words, we need to specify **distance** and the **direction**. Thus we see that distance is a scalar whereas displacement is a vector.

Consider the following two situations:

1. A complete stranger to Zimbabwe is in Harare and wants to travel to Masvingo. Is it sufficient to simply tell them that Masvingo is 350 kilometers from Harare?
2. A well-informed Zimbabwean wants to travel from Harare to Masvingo. Will it be sufficient to tell them that Masvingo is 350 kilometers from Harare?

Clearly, the information in the first case is not sufficient as the stranger would also want to know the direction in which to travel. However, in the second case, it is assumed that the person already has some idea of the location of Masvingo relative to Harare and so specifying **only** distance would suffice.

The following are examples of vectors: force, displacement, acceleration, momentum and velocity . However, the following quantities are scalars and not vectors: area, volume, distance, speed, energy, work, electrical resistance, temperature, mass and time.

2.1 Basic Definitions and Notation

Graphically a vector is represented by an arrow $\overrightarrow{OP}$ defining the direction, the magnitude of the vector being indicated by the length of the arrow. The tail end O of the arrow is called the *origin* or *initial point* of the vector, and the head P is called the *terminal point* or *terminus*. This arrow representing the vector is called a *directed line segment*. The length $|\overrightarrow{OP}|$ is the magnitude of the line segment from O to P .

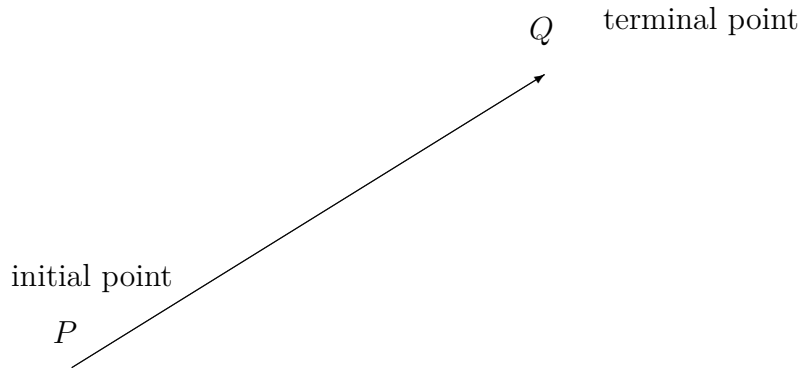


Figure 2.1: Directed line segment

Vectors can be represented in text by bold-case letters, such as **A**, **B**, **C** and so on or lower-case boldface letters such as **a**, **b**, **c** and so on. When written by hand, however, vectors are often denoted by letters with arrows above them, such as $\vec{a}$, $\vec{b}$ and so on or a bar above, such as $\bar{a}$, $\bar{b}$ and so on or a bar below, such as $\underline{a}$, $\underline{b}$ and so on. When the initial point of the vector is fixed, it is called a **fixed** or **localized** vector, otherwise, it is a free vector.

2.2 Unit Vectors

A **unit vector** is a vector of unit length. A unit vector is sometimes denoted by $\hat{e}$ or $\mathbf{\hat{e}}$. Therefore,

$$|\hat{e}| = 1.$$

Any vector can be made into a unit vector by dividing it by its length, that is,

$$\hat{e} = \frac{\mathbf{u}}{|\mathbf{u}|}.$$

So $\frac{\mathbf{u}}{|\mathbf{u}|}$ is a unit vector in the direction of the vector $\mathbf{u}$.

In three-dimensional space $\mathbb{R}^3$, we denote the unit vectors in the positive x -axis, positive y -axis and positive z -axis by $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ respectively. Thus the position vector of a point (x, y, z) is

$$x\mathbf{i} + y\mathbf{j} + z\mathbf{k}.$$

In a similar way, the position vector $\mathbf{r}$ of a point (x, y) in two-dimensional space $\mathbb{R}^2$ is

$$x\mathbf{i} + y\mathbf{j}.$$

In the notation above, the numbers x, y, z are the **components** of the vectors.

We also denote vectors in $\mathbb{R}^3$ and $\mathbb{R}^2$ using **column vectors**. For example the vector $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ in $\mathbb{R}^3$ is denoted by $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and the vector $x\mathbf{i} + y\mathbf{j}$ in $\mathbb{R}^2$ is denoted by $\begin{pmatrix} x \\ y \end{pmatrix}$.

2.3 Magnitude or Length of a Vector

The magnitude or length of a vector $\mathbf{r}$ is denoted by $|\mathbf{r}|$. If a vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, then it can be easily shown by use of Pythagoras' theorem that

$$|\mathbf{r}| = \sqrt{x^2 + y^2 + z^2}.$$

For example if $\mathbf{r} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$, then the magnitude $|\mathbf{r}| = \sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$.

Example 2.3.1. Given $\mathbf{A} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{B} = 2\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}$ and $\mathbf{C} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, find the magnitudes of (i) $\mathbf{C}$, (ii) $\mathbf{A} + \mathbf{B} + \mathbf{C}$ and (iii) $2\mathbf{A} - 2\mathbf{B} - 5\mathbf{C}$.

Solution: (i) $|\mathbf{C}| = |-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}| = \sqrt{(-1)^2 + 2^2 + 2^2} = 3$.
(ii) $\mathbf{A} + \mathbf{B} + \mathbf{C} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k} + 2\mathbf{i} - 4\mathbf{j} - 3\mathbf{k} - \mathbf{i} + 2\mathbf{j} + 2\mathbf{k} = (3 + 2 - 1)\mathbf{i} + (-2 - 4 + 2)\mathbf{j} + (1 - 3 + 2)\mathbf{k} = 4\mathbf{i} - 4\mathbf{j} + 0\mathbf{k}$. Then $|\mathbf{A} + \mathbf{B} + \mathbf{C}| = |4\mathbf{i} - 4\mathbf{j} + 0\mathbf{k}| = \sqrt{4^2 + (-4)^2} = \sqrt{32} = 4\sqrt{2}$.
(iii) $2\mathbf{A} - 3\mathbf{B} - 5\mathbf{C} = 2(3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) - 3(2\mathbf{i} - 4\mathbf{j} - 3\mathbf{k}) - 5(-\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}) = 5\mathbf{i} - 2\mathbf{j} + \mathbf{k}$. Then $|2\mathbf{A} - 3\mathbf{B} - 5\mathbf{C}| = |5\mathbf{i} - 2\mathbf{j} + \mathbf{k}| = \sqrt{5^2 + (-2)^2 + 1^2} = \sqrt{30}$.

Example 2.3.2. Find the component form and magnitude of the vector $\mathbf{A}$ having initial point $(-2, 3, 1)$ and terminal point $(0, -4, 4)$. Then find a unit vector in the direction of $\mathbf{A}$.

Solution: The component form of $\mathbf{A}$ is

$$\mathbf{A} = (0 - (-2), -4 - 3, 4 - 1) = (2, -7, 3)$$

which implies that its magnitude is

$$|\mathbf{A}| = \sqrt{2^2 + (-7)^2 + 3^2} = \sqrt{62}.$$

The unit vector in the direction of $\mathbf{A}$ is

$$\mathbf{U} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{1}{\sqrt{62}}(2, -7, 3) = \left(\frac{2}{\sqrt{62}}, \frac{-7}{\sqrt{62}}, \frac{3}{\sqrt{62}} \right).$$

2.4 Parallel Vectors

Two non-zero vectors $\mathbf{A}$ and $\mathbf{B}$ are **parallel** if there is some scalar c such that $\mathbf{A} = c\mathbf{B}$.

Example 2.4.1. Vector $\mathbf{A}$ has initial point $(2, -1, 3)$ and terminal point $(-4, 7, 5)$. Which of the following vectors is parallel to $\mathbf{A}$? (i) $\mathbf{B} = (3, -4, -1)$ and (ii) $\mathbf{C} = (12, -16, 4)$.

Solution: Writing $\mathbf{A}$ in component form

$$\mathbf{A} = (-4 - 2, 7 - (-1), 5 - 3) = (-6, 8, 2).$$

(i) Because $\mathbf{B} = (3, -4, -1) = -\frac{1}{2}(-6, 8, 2) = -\frac{1}{2}\mathbf{A}$, then $\mathbf{B}$ is parallel to $\mathbf{A}$.

(ii) In this case, you want to find a scalar c such that

$$\begin{aligned}(12, -16, 4) &= c(-6, 8, 2) \\ 12 &= -6c \Rightarrow c = -2 \\ -16 &= 8c \Rightarrow c = -2 \\ 4 &= 2c \Rightarrow c = 2.\end{aligned}$$

Because there is no c for which the equation has a solution, the vectors are not parallel.

Definition

Two or more vectors are said to be collinear vectors, when they are along the same lines or parallel lines.

Theorem 2.4.1. Let $\mathbf{a}$ and $\mathbf{b}$ be non-zero and non-collinear vectors. Then $x\mathbf{a} + y\mathbf{b} = \mathbf{0}$ implies that $x = y = 0$.

Proof. Suppose $x\mathbf{a} + y\mathbf{b} = \mathbf{0}$ where $x \neq 0$. This means that $\mathbf{a} = -(\frac{y}{x})\mathbf{b}$. Thus the vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel. In other words they are parallel to the same line or are collinear. Contradiction. Hence x must be equal to zero and so $y\mathbf{b} = \mathbf{0}$. Therefore $y = 0$ as $\mathbf{b} \neq \mathbf{0}$. $\square$

Theorem 2.4.2. Let $\mathbf{a}$ and $\mathbf{b}$ be non-zero and non-collinear vectors. Then $x_1\mathbf{a} + y_1\mathbf{b} = x_2\mathbf{a} + y_2\mathbf{b}$ implies that $x_1 = x_2$ and $y_1 = y_2$.

The proof of this theorem is left as an exercise for you.

2.5 Vector Algebra

The operations of addition, subtraction and multiplication familiar in the algebra of numbers or scalars are, with suitable definition, capable of extension to an algebra of vectors. The following definitions are fundamental.

- (i) Two vectors **A** and **B** are *equal* if they have the same magnitude and direction regardless of the position of their initial points.
- (ii) A vector having direction opposite to that of vector **A** but having the same magnitude is denoted by **-A**.
- (iii) The **sum** or **resultant** of vectors **A** and **B** is a vector **C** formed by placing the initial point of **B** on the terminal point of **A** and then joining the initial point of **A** to the terminal point of **B**. This sum is written $\mathbf{A} + \mathbf{B}$, i.e. $\mathbf{C} = \mathbf{A} + \mathbf{B}$.

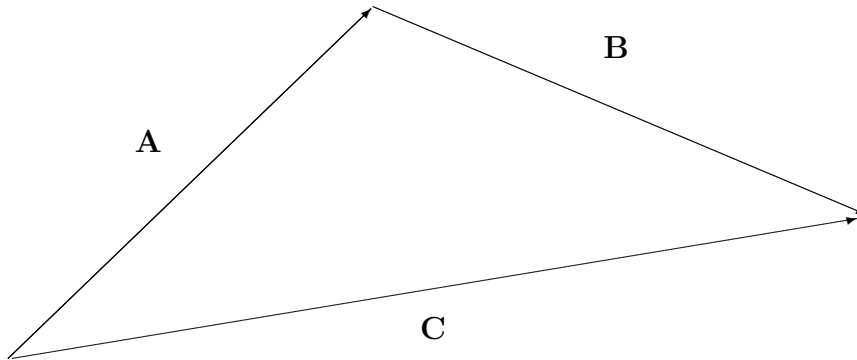


Figure 2.2: $\mathbf{A} + \mathbf{B} = \mathbf{C}$

Theorem 2.5.1. *Vector addition is commutative, that is*

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A},$$

*for each pair of vectors **A** and **B**.*

Theorem 2.5.2. *Vector addition is associative, that is*

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C},$$

*for any triple of vectors **A**, **B** and **C**.*

- (iv) The **difference** of vectors **A** and **B**, represented by $\mathbf{A} - \mathbf{B}$, is that vector **C** which added to **B** yields vector **A**. Equivalently, $\mathbf{A} - \mathbf{B}$ can be defined as the sum $\mathbf{A} + (-\mathbf{B})$.

If $\mathbf{A} = \mathbf{B}$, then $\mathbf{A} - \mathbf{B}$ is defined as the **null** or **zero vector** and is represented by the symbol **0** or simply 0. It has zero magnitude and no specific direction. A vector which is not null is a **proper vector**. All vectors will be assumed proper unless otherwise stated.

- (v) The **product** of a vector **A** by a scalar m is a vector $m\mathbf{A}$ with magnitude $|m|$ times the magnitude of **A** and with direction the same as or opposite to that of **A**, according as m is positive or negative. If $m = 0$, $m\mathbf{A}$ is the null vector.

2.6 Laws of Vector Algebra

If $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are vectors and m and n are scalars, then

- (i) $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ Commutative Law for Addition.
- (ii) $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$ Associative Law of Addition.
- (iii) $m(n\mathbf{A}) = mn\mathbf{A} = n(m\mathbf{A})$ Associative Law for Multiplication.
- (iv) $(m + n)\mathbf{A} = m\mathbf{A} + n\mathbf{A}$ Distributive Law.
- (v) $m(\mathbf{A} + \mathbf{B}) = m\mathbf{A} + m\mathbf{B}$ Distributive Law.

2.7 The Dot or Scalar Product

So far we have studied two operations with vectors, vector addition and multiplication by a scalar, each of which yield another vector. In this section you will study a third vector operation, called the **dot product**, this product yields a scalar, rather than a vector.

The **dot** or **scalar product** of two vectors $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, denoted by $\mathbf{A} \cdot \mathbf{B}$ (read $\mathbf{A}$ dot $\mathbf{B}$) is defined as the product of the magnitudes of $\mathbf{A}$ and $\mathbf{B}$ and the cosine of the angle θ between them, that is,

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos \theta = A_1B_1 + A_2B_2 + A_3B_3$$

where $0 \leq \theta \leq \pi$.

Example 2.7.1. Given $\mathbf{A} = 2\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. Compute the scalar product of $\mathbf{A}$ and $\mathbf{B}$.

Solution: From the definition, the scalar product is given by

$$\mathbf{A} \cdot \mathbf{B} = (2) \cdot (1) + (4) \cdot (-3) + (6)(2) = 2 - 12 + 12 = 2.$$

The following laws are valid

- (i) $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$ Commutative Law for Dot Products.
- (ii) $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$ Distributive Law.
- (iii) $m(\mathbf{A} \cdot \mathbf{B}) = (m\mathbf{A}) \cdot \mathbf{B} = \mathbf{A} \cdot (m\mathbf{B}) = (\mathbf{A} \cdot \mathbf{B})m$, where m is a scalar.
- (iv) $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$, $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$.
- (v) If $\mathbf{A} \cdot \mathbf{B} = 0$ and $\mathbf{A}$ and $\mathbf{B}$ are not null vectors, then $\mathbf{A}$ and $\mathbf{B}$ are perpendicular.

2.7.1 Angle Between Two Vectors

If θ is the angle between two non-zero vectors $\mathbf{A}$ and $\mathbf{B}$, then

$$\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|}.$$

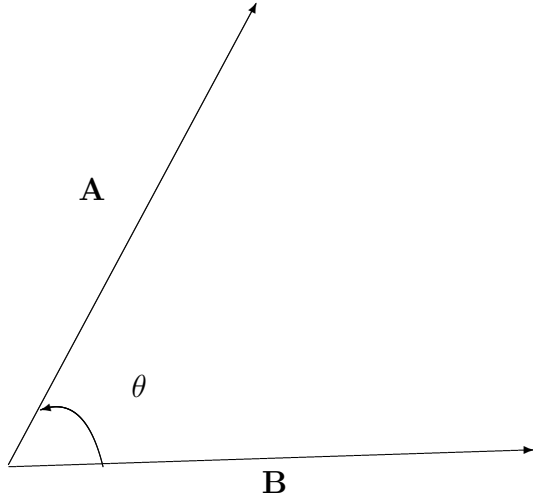


Figure 2.3: The angle between two vectors

2.7.2 Definition of Orthogonal Vectors

The vectors $\mathbf{A}$ and $\mathbf{B}$ are **orthogonal** if $\mathbf{A} \cdot \mathbf{B} = 0$. Two non-zero vectors are orthogonal if and only if the angle between them is $\theta = \frac{\pi}{2}$.

Example 2.7.2. For $\mathbf{A} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$, $\mathbf{B} = -4\mathbf{i} + 2\mathbf{k}$, $\mathbf{C} = \mathbf{i} - \mathbf{j} - 2\mathbf{k}$ and $\mathbf{D} = 2\mathbf{i} - \mathbf{k}$, find the angles between the following pairs of vectors. (i) $\mathbf{A}$ and $\mathbf{B}$ (ii) $\mathbf{A}$ and $\mathbf{C}$ (iii) $\mathbf{B}$ and $\mathbf{D}$.

Solution:

$$(i) \cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{A}||\mathbf{B}|} = \frac{-12 + 4}{\sqrt{14}\sqrt{20}} = \frac{-8}{2\sqrt{14}\sqrt{5}} = \frac{-4}{\sqrt{70}}. \text{ Because } \mathbf{A} \cdot \mathbf{B} < 0,$$

$$\theta = \cos^{-1} \left(\frac{-4}{\sqrt{70}} \right) = 2.069 \text{ radians.}$$

$$(ii) \cos \theta = \frac{\mathbf{A} \cdot \mathbf{C}}{|\mathbf{A}||\mathbf{C}|} = \frac{3 + 1 - 4}{\sqrt{14}\sqrt{6}} = \frac{0}{\sqrt{84}} = 0. \text{ Because } \mathbf{A} \cdot \mathbf{C} = 0, \mathbf{A} \text{ and } \mathbf{C} \text{ are orthogonal.}$$

Furthermore, $\theta = \frac{\pi}{2}$.

$$(iii) \cos \theta = \frac{\mathbf{B} \cdot \mathbf{D}}{|\mathbf{B}||\mathbf{D}|} = \frac{-8 + 0 - 2}{\sqrt{20}\sqrt{5}} = \frac{-10}{\sqrt{100}} = -1. \text{ Consequently, } \theta = \pi.$$

Exercise

Prove that a parallelogram $ABCD$ is a rhombus if and only if its diagonals are orthogonal.

2.8 The Cross or Vector Product

Many applications in physics, engineering and geometry involve finding a vector in space that is orthogonal to two given vectors. In this section we will study a product that will yield such a vector. The **cross** or **vector product** of $\mathbf{A}$ and $\mathbf{B}$ is a vector $\mathbf{C} = \mathbf{A} \times \mathbf{B}$ (read $\mathbf{A}$ cross $\mathbf{B}$),

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin \theta \mathbf{n},$$

where θ is the angle between the vectors, and the unit vector $\mathbf{n}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$, with $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{n}$ forming a right-handed system.

The following laws are valid.

- (i) $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$ Anticommutative Law.
- (ii) $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = \mathbf{A} \times \mathbf{B} + \mathbf{A} \times \mathbf{C}$ Distributive Law.
- (iii) $m(\mathbf{A} \times \mathbf{B}) = (m\mathbf{A}) \times \mathbf{B} = \mathbf{A} \times (m\mathbf{B}) = (\mathbf{A} \times \mathbf{B})m$, where m is a scalar.
- (iv) $\mathbf{i} \times \mathbf{i} = \mathbf{j} \times \mathbf{j} = \mathbf{k} \times \mathbf{k} = 0$, $\mathbf{i} \times \mathbf{j} = \mathbf{k}$, $\mathbf{j} \times \mathbf{k} = \mathbf{i}$, $\mathbf{k} \times \mathbf{i} = \mathbf{j}$.
- (v) If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$ and $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$, then

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \end{vmatrix}.$$

- (vi) The magnitude of $\mathbf{A} \times \mathbf{B}$ is the same as the area of a parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$.
- (vii) If $\mathbf{A} \times \mathbf{B} = 0$ and $\mathbf{A}$ and $\mathbf{B}$ are not null vectors, then $\mathbf{A}$ and $\mathbf{B}$ are parallel.

Example 2.8.1. Given $\mathbf{A} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{B} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$, find the following. (i) $\mathbf{A} \times \mathbf{B}$
(ii) $\mathbf{B} \times \mathbf{A}$ (iii) $\mathbf{B} \times \mathbf{B}$.

Solution:

(i)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 1 \\ 3 & 1 & -2 \end{vmatrix} = \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 1 & 1 \\ 3 & -2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} \mathbf{k} = 3\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}.$$

(ii)

$$\mathbf{B} \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -2 \\ 1 & -2 & 1 \end{vmatrix} = \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix} \mathbf{k} = -3\mathbf{i} - 5\mathbf{j} - 7\mathbf{k}.$$

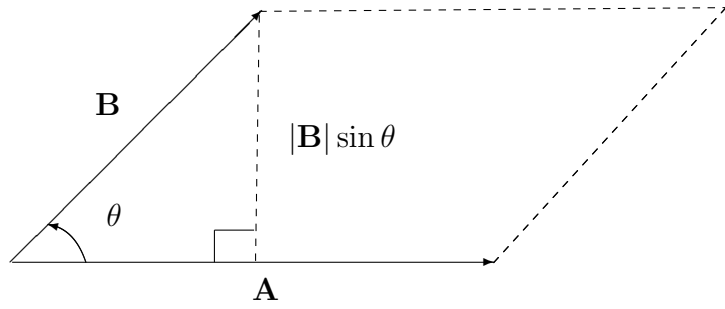


Figure 2.4: The vectors $\mathbf{A}$ and $\mathbf{B}$ form adjacent sides of a parallelogram

(iii)

$$\mathbf{B} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 1 & -2 \\ 3 & 1 & -2 \end{vmatrix} = 0.$$

Example 2.8.2. Find the area of the parallelogram determined by $\mathbf{A} = \mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = -6\mathbf{j} + 5\mathbf{k}$.

Solution:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & -3 \\ 0 & -6 & 5 \end{vmatrix} = -13\mathbf{i} - 5\mathbf{j} - 6\mathbf{k}.$$

Therefore

$$|\mathbf{A} \times \mathbf{B}| = \sqrt{(-13)^2 + 5^2 + 6^2} = \sqrt{230}$$

which is the desired area.

Example 2.8.3. Find a unit vector that is orthogonal to both $\mathbf{A} = \mathbf{i} - 4\mathbf{j} + \mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} + 3\mathbf{j}$.

Solution: The cross product $\mathbf{A} \times \mathbf{B}$ is orthogonal to both $\mathbf{A}$ and $\mathbf{B}$.

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -4 & 1 \\ 2 & 3 & 0 \end{vmatrix} = -3\mathbf{i} + 2\mathbf{j} + 11\mathbf{k}.$$

Because $|\mathbf{A} \times \mathbf{B}| = \sqrt{(-3)^2 + 2^2 + 11^2} = \sqrt{134}$, a unit vector orthogonal to both $\mathbf{A}$ and $\mathbf{B}$ is

$$\frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} = -\frac{3}{\sqrt{134}}\mathbf{i} + \frac{2}{\sqrt{134}}\mathbf{j} + \frac{11}{\sqrt{134}}\mathbf{k}.$$

2.9 The Scalar Triple Product

For vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ in space, the dot product of $\mathbf{A}$ and $\mathbf{B} \times \mathbf{C}$

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$$

is called the **scalar triple product**. The following laws are valid.

- (i) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) = \text{volume of a parallelopiped having } \mathbf{A}, \mathbf{B} \text{ and } \mathbf{C} \text{ as edges.}$

If $\mathbf{A} = A_1\mathbf{i} + A_2\mathbf{j} + A_3\mathbf{k}$, $\mathbf{B} = B_1\mathbf{i} + B_2\mathbf{j} + B_3\mathbf{k}$ and $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$, then

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix}.$$

- (ii) As a consequence, the volume of the parallelopiped is 0 if and only if the three vectors are **coplanar**. That is, if the vectors $\mathbf{A} = (A_1, A_2, A_3)$, $\mathbf{B} = (B_1, B_2, B_3)$ and $\mathbf{C} = (C_1, C_2, C_3)$ have the same initial point, then they lie in the same plane if and only if

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} A_1 & A_2 & A_3 \\ B_1 & B_2 & B_3 \\ C_1 & C_2 & C_3 \end{vmatrix} = 0.$$

Example 2.9.1. Find the volume of the parallelopiped having $\mathbf{A} = 3\mathbf{i} - 5\mathbf{j} + \mathbf{k}$, $\mathbf{B} = 2\mathbf{j} - 2\mathbf{k}$ and $\mathbf{C} = 3\mathbf{i} + \mathbf{j} + \mathbf{k}$ as adjacent edges.

Solution:

$$V = |\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})| = \begin{vmatrix} 3 & -5 & 1 \\ 0 & 2 & -2 \\ 3 & 1 & 1 \end{vmatrix} = 3 \begin{vmatrix} 2 & -2 \\ 1 & 1 \end{vmatrix} - (-5) \begin{vmatrix} 0 & -2 \\ 3 & 1 \end{vmatrix} + (1) \begin{vmatrix} 0 & 2 \\ 3 & 1 \end{vmatrix} = 3(4) + 5(6) + 1(-6) = 36.$$

Example 2.9.2. Determine whether the four points $A(-2, 0, 3)$, $B(1, 0, 0)$, $C(1, -3, 3)$ and $D(4, 1, -2)$ are coplanar.

Solution: We construct three vectors from the four points, $\mathbf{a} = \overrightarrow{AD} = (6, 1, -5)$, $\mathbf{b} = \overrightarrow{AB} = (3, 0, -3)$, $\mathbf{c} = \overrightarrow{AC} = (3, -3, 0)$. The scalar product is

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} 6 & 1 & -5 \\ 3 & 0 & -3 \\ 3 & -3 & 0 \end{vmatrix} = 6(-9) - (1)(9) + (-5)(-9) = -18 \neq 0.$$

Hence not coplanar.

2.10 The Vector Triple Product

Let $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ be a triple of vectors. Then the vector

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$$

is called the **triple vector product** of $\mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ in that order. The evaluation of a vector triple product can be made easier using the vector identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}.$$

Example 2.10.1. Given the vectors $\mathbf{A} = \mathbf{i} + 3\mathbf{j} - \mathbf{k}$, $\mathbf{B} = -2\mathbf{i} + \mathbf{j} - 5\mathbf{k}$ and $\mathbf{C} = 3\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$. Verify the vector identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}.$$

Solution: Starting with the left hand side

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 1 & -5 \\ 3 & -2 & 7 \end{vmatrix} = -3\mathbf{i} - \mathbf{j} + \mathbf{k}.$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 3 & -1 \\ -3 & -1 & 1 \end{vmatrix} = 2\mathbf{i} + 2\mathbf{j} + 8\mathbf{k}.$$

The right hand side gives

$$(\mathbf{A} \cdot \mathbf{C}) = 3(1) + 3(-2) + 7(-1) = -10, (\mathbf{A} \cdot \mathbf{C})\mathbf{B} = -10(-2\mathbf{i} + \mathbf{j} - 5\mathbf{k}) = 20\mathbf{i} - 10\mathbf{j} + 50\mathbf{k}.$$

$$(\mathbf{A} \cdot \mathbf{B}) = (-2)(1) + (3)(1) + (-5)(-1) = 6, (\mathbf{A} \cdot \mathbf{B})\mathbf{C} = 6(3\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}) = 18\mathbf{i} - 12\mathbf{j} + 42\mathbf{k}.$$

Then

$$(\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C} = (20 - 18)\mathbf{i} + (-10 + 12)\mathbf{j} + (50 - 42)\mathbf{k} = 2\mathbf{i} + 2\mathbf{j} + 8\mathbf{k}.$$

2.11 Tutorial Questions

- Find a, b and c if $(a + b - c)\mathbf{i} + (c - 1)\mathbf{j} + (a + c)\mathbf{k} = \mathbf{0}$.
- Which of the following are unit vectors?
(i) $\mathbf{i}$ (ii) $\mathbf{i} + \frac{1}{2}\mathbf{j}$ (iii) $-\mathbf{j} + \mathbf{k}$ (iv) $\frac{1}{\sqrt{2}}\mathbf{i} + \frac{1}{\sqrt{2}}\mathbf{j}$.
- If $|\mathbf{A}| = 3$. What is $|4\mathbf{A}|$?
- Determine the unit vector having the same direction as $4\mathbf{i} + 3\mathbf{j}$.
- Let $\mathbf{a}$ and $\mathbf{b}$ be non-zero and non-collinear vectors. Prove that if $x_1\mathbf{a} + y_1\mathbf{b} = x_2\mathbf{a} + y_2\mathbf{b}$ then $x_1 = x_2$ and $y_1 = y_2$.
- Find the scalar product of $3\mathbf{i} + 8\mathbf{j} - 2\mathbf{k}$ and $5\mathbf{i} + \mathbf{j} + 2\mathbf{k}$.
- Find the angle between $2\mathbf{i}$ and $3\mathbf{i} + 4\mathbf{j}$.
- Prove that a parallelogram $ABCD$ is a rhombus if and only if its diagonals are perpendicular bisectors of each other.
- If $\mathbf{A} = \mathbf{i} - \mathbf{j}$ and $\mathbf{B} = -\mathbf{j} + 2\mathbf{k}$, show that $(\mathbf{A} + \mathbf{B}) \cdot (\mathbf{A} - 2\mathbf{B}) = -9$.
- Find $\mathbf{A} \times \mathbf{B}$ if
(i) $\mathbf{A} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{B} = \mathbf{i} + \mathbf{j} - 4\mathbf{k}$ (ii) $\mathbf{A} = 2\mathbf{i} + \mathbf{j} + 7\mathbf{k}$ and $\mathbf{B} = 3\mathbf{i} + \mathbf{j} - \mathbf{k}$
(iii) $\mathbf{A} = 3\mathbf{i}$ and $\mathbf{B} = \mathbf{j}$ (iv) $\mathbf{A} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{B} = 2\mathbf{i} + \mathbf{j} - \mathbf{k}$.

11. Find a unit vector perpendicular to both $\mathbf{A} = 3\mathbf{i} + \mathbf{j}$ and $\mathbf{B} = 2\mathbf{i} - \mathbf{j} - 5\mathbf{k}$.
12. If $\mathbf{A}$ and $\mathbf{B}$ are unit vectors, show that $|\mathbf{A} \times \mathbf{B}|^2 = 1 - (\mathbf{A} \cdot \mathbf{B})^2$.
13. Show that $\mathbf{A} = (\cos \theta, \sin \theta \cos \phi, \sin \theta \sin \phi)$ is a unit vector.
14. If $\mathbf{A} = 3\mathbf{i} - \mathbf{j} - 4\mathbf{k}$, $\mathbf{B} = -2\mathbf{i} + 4\mathbf{j} - 3\mathbf{k}$, $\mathbf{C} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$, find (i) $2\mathbf{A} - \mathbf{B} + 3\mathbf{C}$ (ii) $|\mathbf{A} + \mathbf{B} + \mathbf{C}|$
(iii) $|3\mathbf{A} - 2\mathbf{B} + 4\mathbf{C}|$ (iv) a unit vector parallel to $3\mathbf{A} - 2\mathbf{B} + 4\mathbf{C}$.
15. Determine the value of a so that $\mathbf{A} = 2\mathbf{i} + a\mathbf{j} + \mathbf{k}$ and $\mathbf{B} = 4\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ are perpendicular.
16. Show that the vectors $\mathbf{A} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{B} = \mathbf{i} - 3\mathbf{j} + 5\mathbf{k}$ and $\mathbf{C} = 2\mathbf{i} + \mathbf{j} - 4\mathbf{k}$ form a right angled triangle.
17. Show that $|\mathbf{A} \times \mathbf{B}|^2 + (\mathbf{A} \cdot \mathbf{B})^2 = |\mathbf{A}|^2 |\mathbf{B}|^2$.
18. Find the area of a parallelogram with edges $\mathbf{A} = 3\mathbf{i} + \mathbf{j} - 2\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$.
19. Find the area of the triangle having vertices at $P(1, 3, 2)$, $Q(2, -1, 1)$ and $R(-1, 2, 3)$.
20. If $\mathbf{A} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ and $\mathbf{B} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, find a vector of magnitude 5 perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.
21. Find the volume of the parallelepiped whose edges are represented by $\mathbf{A} = 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$,
 $\mathbf{B} = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\mathbf{C} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.
22. Determine whether the four points $A(-2, 0, 3)$, $B(1, 0, 0)$, $C(1, -3, 3)$ and $D(4, 1, -2)$ are coplanar.
23. Prove that $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) + \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = 0$.